ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MATHEMATICS

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 2. Numbers and algebra (25), functions and graphs (15), financial mathematics (15), measurement and trigonometry (25), data handling and probability (20).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations, diagrams and graphs that you used to find your answers. Answers only will not necessarily earn full marks.
4. An approved scientific calculator (non-programmable and non-graphical) may be used, unless stated otherwise.
5. Round off answers to TWO decimal places, unless stated otherwise.
6. Diagrams are not necessarily drawn to scale.
7. Write neatly and legibly.

QUESTION 1: NUMBERS AND ALGEBRA

1.1 Simplify without using a calculator:

1.1.1 $(3x^2y^3)^2 \div 9x^3y$ (3)

1.1.2 $\dfrac{2^{x+2} - 2^x}{3 \cdot 2^x}$ (3)

1.2 Write 0,000 456 in scientific notation. (1)

1.3 Factorise fully:

1.3.1 $6x^2 - 15x$ (1)

1.3.2 $x^2 - 7x + 12$ (2)

1.3.3 $4x^2 - 25$ (2)

1.4 Simplify: $\dfrac{x^2 - 9}{x^2 + x - 6}$ (3)

1.5 Solve for x:

1.5.1 $3(x - 2) = 2x + 5$ (2)

1.5.2 $x^2 - 5x - 14 = 0$ (2)

1.5.3 $5 - 2x > 11$ (2)

1.6 Solve simultaneously: $x + y = 7$ and $2x - y = 5$ (2)

1.7 Make r the subject of the formula $V = \pi r^2 h$. (2)

[25]

QUESTION 2: FUNCTIONS AND GRAPHS

2.1 Given: $f(x) = 2x - 4$ and $g(x) = -x^2 + 4$

2.1.1 Determine the x- and y-intercepts of f. (2)

2.1.2 Determine the x- and y-intercepts and the turning point of g. (3)

2.1.3 Sketch the graphs of f and g on the same set of axes. Show all intercepts and the turning point. (4)

2.1.4 Calculate the coordinates of the points where the graphs of f and g intersect. (4)

2.1.5 For which values of x is $g(x) \geq f(x)$? (2)

[15]

QUESTION 3: FINANCIAL MATHEMATICS

3.1 Thabo invests R12 000 for 3 years. Calculate the final amount if he receives:

3.1.1 8% per annum simple interest (3)

3.1.2 8% per annum interest compounded annually (3)

3.1.3 How much more does he receive with compound interest? (1)

3.2 A cellphone costs R6 000 cash. Naledi buys it on hire purchase: she pays a 10% deposit, and the balance plus simple interest of 15% per annum is paid off in equal monthly instalments over 2 years. Calculate her monthly instalment. (5)

3.3 A loaf of bread costs R20 today. If inflation is 6% per annum, what will it cost in 2 years' time? (2)

3.4 If \$1 = R18,50, how many rand will a student need to buy a \$250 laptop online? (1)

[15]

QUESTION 4: MEASUREMENT AND TRIGONOMETRY

4.1 In right-angled triangle ABC, $\hat{B} = 90°$, AB = 8 cm and BC = 15 cm.

4.1.1 Calculate the length of AC. (2)

4.1.2 Write down the value of sin A, cos A and tan A as fractions. (3)

4.1.3 Calculate the size of $\hat{A}$. (2)

4.2 A 6 m ladder leans against a vertical wall and makes an angle of 70° with the level ground. Calculate:

4.2.1 how high up the wall the ladder reaches (2)

4.2.2 how far the foot of the ladder is from the wall (2)

4.3 A closed cylindrical water tank has a diameter of 1,4 m and a height of 2 m. Use the π button on your calculator.

4.3.1 Calculate the volume of the tank in m^3. (3)

4.3.2 How many litres of water can the tank hold? ($1\ m^3 = 1\ 000\ \ell$) (1)

4.3.3 Calculate the total outer surface area of the tank. (3)

4.4 A rectangular floor measuring 4,5 m by 3,2 m must be tiled with square tiles of 40 cm × 40 cm. The tiles are sold in boxes of 12.

4.4.1 Calculate the number of tiles needed. (3)

4.4.2 How many boxes must be bought? (1)

4.5 Convert: (a) 3 500 mm to m (b) 2,4 m^2 to cm^2 (c) 90 km/h to m/s (3)

[25]

QUESTION 5: DATA HANDLING AND PROBABILITY

5.1 The test marks (out of 100) of ten NC(V) students are: 45 62 58 71 62 80 39 62 55 66

5.1.1 Arrange the marks in ascending order. (1)

5.1.2 Calculate the mean. (2)

5.1.3 Determine the median and the mode. (2)

5.1.4 Calculate the range. (1)

5.1.5 Which measure of central tendency best describes these marks? Give a reason. (2)

5.1.6 What percentage of the students scored 60 or more? (2)

5.2 A box contains 5 red, 3 blue and 2 green fuses of the same size. One fuse is taken out without looking. Determine:

5.2.1 P(blue) (1)

5.2.2 P(not green) (2)

5.2.3 P(yellow) (1)

5.3 The table shows how 40 students travel to college.

Transport	**Taxi**	**Bus**	**Walk**	**Car**
Number of students	18	8	10	4

5.3.1 Draw a bar graph to represent the data. (4)

5.3.2 If a pie chart is drawn, calculate the angle of the sector for "Taxi". (2)

[20]

TOTAL: 100

END OF QUESTION PAPER